

Student Performance Analyzer

Internship Project Report

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Objective

To analyze student performance data and identify factors influencing academic outcomes.

Dataset Description

The dataset consists of 1000 student records containing demographic details, parental education levels, lunch types, test preparation status, and scores in mathematics, reading, and writing.

Methodology

The project involved data exploration, missing value analysis, duplicate checking, descriptive statistical analysis, feature engineering, and visualization using Python libraries such as Pandas, Matplotlib, and Seaborn.

Key Findings

Gender Analysis

Male students performed slightly better in mathematics, whereas female students demonstrated stronger performance in reading and writing.

Correlation Analysis

Strong positive correlations were observed among all subjects. The highest correlation existed between reading and writing scores ($r = 0.95$).

Test Preparation Analysis

Students who completed the test preparation course achieved higher average scores compared to those who did not complete the course.

Lunch Analysis

Students receiving standard lunch demonstrated better academic performance than those receiving free or reduced lunch.

Top Students Analysis

Top-performing students maintained consistently excellent performance across all subjects.

Conclusion

The analysis indicates that academic performance is influenced by multiple factors, including preparation and socioeconomic indicators. The findings provide valuable educational insights that can support educational decision-making.